

Introduction

This documentation explores IP addressing and subnet, while also explaining some concepts revolving around IP addressing and subnetting. First, we will explore what IP addresses are and why they are important in networking. Secondly, we will explore the concept of subnetting and some terminology around subnetting. Lastly, we will explore these concepts through a simulated network topology.

What is an IP address?

We rely on different protocols that help end-hosts, such as users, to use the services they rely on. IP is one of these protocols.

IP stands for Internet Protocol, and is a 32-bit logical address. For regular people, we understand words such as “hello”, because we understand what each character means. However, computers are the opposite to humans. They don't understand this, but understand something called binary. To better understand binary, we need to understand what binary is?

You may think Binary is something highly complex, well, Binary is a bunch of 0's and 1's string together to perform some operation, for a digital device. An example of an IP address is: 192.168.0.0/24.

IP addresses support 2 versions:

- IPv4 - Current version of IP address being used
- IPv6 - The next generation of IP addresses being used

IPv4 is the current addressing scheme we use, but we are slowly migrating to IPv6 due the limited available IPv4 addresses available. .

What is a subnet mask?

Going back to our IP address: 192.168.0.0/24. The /24 is a prefix mask that lets the router know how many binary digits to read up until. Depending on the class of the IPv4 addresses, the router know which octets to ignore:

- Class A - /8 - The 2nd,3rd,4th octet is ignored.
- Class B - /16 - The 3rd,4th octet is ignored.
- Class C - /24 - The 4th octet is ignored.

Key Subnetting Values:

This part will focus on the idea of classful addressing, and not CIDR notation, which is outside the scope.

So, we understand subnetting revolves around the idea of having a large IP address, and creating smaller networks, using the same IP address. But, there are a few terms we need to understand for subnetting:

Network Address:

This denotes the start of the subnet. For example, Lets say we have the following IP address:

192.168.1.0/24

Binary = 11000000.10101000.00000001.00000000
 192 .168 .1 .0

We switch each bit in the last octet to 0, we get our network address.

Broadcast Address:

This denotes the end of the subnet. For example, Lets say we have the following IP address:

192.168.1.0 /24

Binary = 11000000.10101000.00000001.11111111
 192 .168 .1 .255

Broadcast = 192.168.1.255 /24

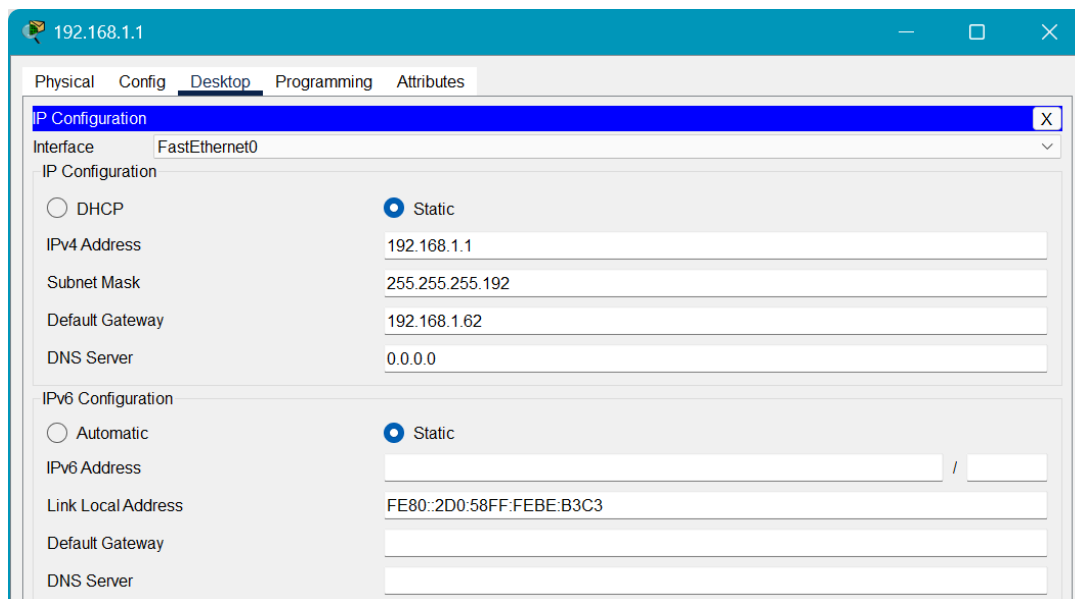
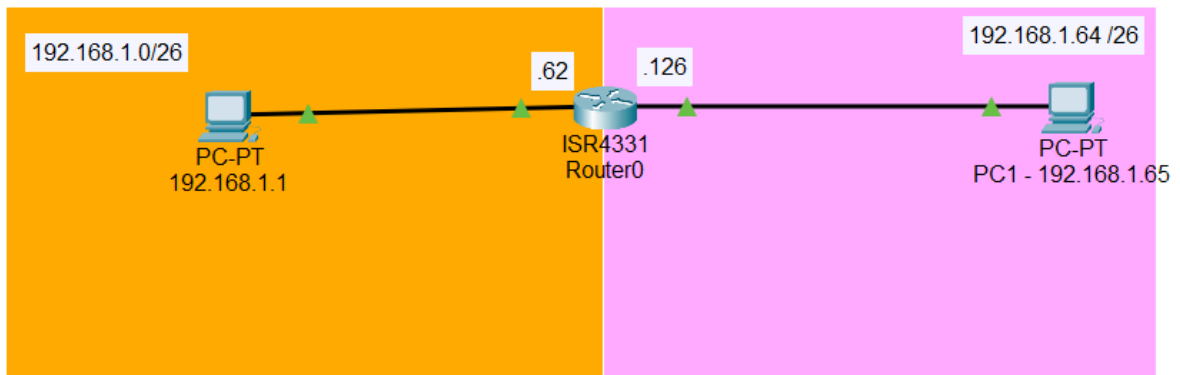
Usable Host Range

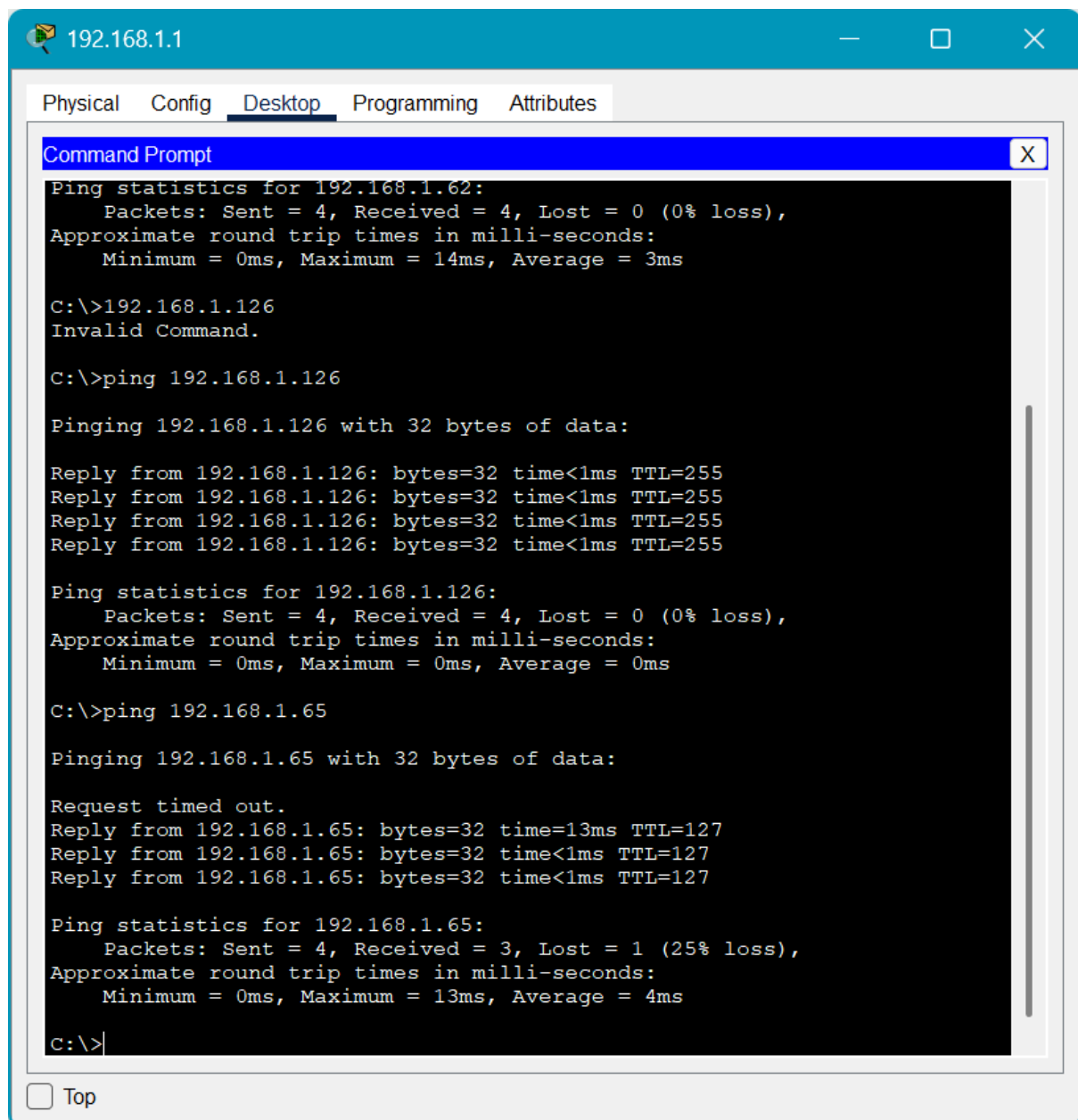
This lets us know how many IP address are available to use within the subnet. This is typically, the 1st IP address after the network address, and the last IP address before the Broadcast address.

So, using the same IP address: 192.168.1.0 /24, What would you think our host range would be?

Network Address	Broadcast Address
(192.168.1.0)	(192.168.1.255)
192.168.1.1 - 192.168.1.254	

Network Topology





These images are screenshots from Packet Tracer

Subnetting in Practice

The IP address and mask I have used is: 192.168.1.0 /26

This means, I am not using the full 24 bits to create 1 large network, but changing some of the bits within the last octet to make smaller networks.

	/25	/26	/27	/28	/29	/30	/31	/32
Binary to the power of ()	2x(8)	2x(7)	2x(6)	2x(5)	2x(4)	2x(3)	2x(2)	2x(1)
value	128	64	32	16	8	4	2	1
Bits	-	1	0	0	0	0	0	0

So, from the 27th bit onwards, I can do anything I want, which we can see in the table above. We can also infer that each subnet is going to contain 64 addresses per subnet due the mask:

- 192.168.1.0 /26
- 192.168.1.64 /26
- 192.168.1.128 /26
- 192.168.1.192 /26

Now what does this mean exactly?

So we used the following subnets:

- 192.168.1.0 /26
 - Network Address: 192.168.1.0
 - Broadcast Address: 192.168.1.63
 - 1st usable host: 192.168.1.1
 - Last usable host: 192.168.1.62
- 192.168.1.64 /26
 - Network Address: 192.168.1.64
 - Broadcast Address: 192.168.1.127
 - 1st usable host: 192.168.1.65
 - Last usable host: 192.168.1.126

Looking at the topology above, we have a router that has separated two large networks into two separate networks. Lets say you have two departments sharing the same IP addresses from the one big IP address of 192.168.1.0 /24. Let's say the network is being over-utilised by one department. This would affect all the users because they are sharing the same network.

Now let's say we have split that one large network into a smaller network. Now we can control the performance of the network at a granular level since we can make changes that only affect one department, and does not compromise the other department.

Key Takeaway:

In this documentation, we explored the fundamentals of IP addressing and subnetting. We learned that an IPv4 address is a 32-bit logical address, split into 4 octets, and that every device on a network requires a unique IP address to communicate.

We explored how a subnet mask defines the network and host portions of an IP address, and how CIDR notation such as /26 allows us to create smaller, more efficient networks from a larger address space. Using /26 gives us 64 addresses per subnet, of which 62 are usable — with the network address and broadcast address reserved at either end.

Through our Packet Tracer topology, we demonstrated how 192.168.1.0/24 can be divided into four /26 subnets, and configured two of them — 192.168.1.0/26 and 192.168.1.64/26 — connected via a router. We verified connectivity between the two subnets using a successful ping.

Understanding subnetting is a foundational skill in networking, underpinning routing decisions, firewall rules, and ACL configuration — all of which are critical in cybersecurity roles.